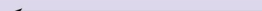


TABLE I

	LF	HF	UHF	Microwave
Frequency range	<135 kHz	13.56 MHz	860-930 MHz	2.45 GHz
Standards specifications	ISO/IEC 18000-2	ISO/IEC 18000-3, AutoID HF class 1, ISO 15693, ISO 14443 (A/B)	ISO/IEC 18000-6, Auto ID class 0, class 1	ISO/IEC 18000-4
General	Larger antennae resulting in higher-cost tags, least susceptible to performance degradations from metals and liquids	Less expensive than LF tags. Best suited for applications that do not require long-range reading of a large number of tags. This frequency has the widest application scope	In volume UHF tags have the potential to be cheaper than LF or HF due to recent advances in IC design. Good for reading multiple tags at long range. More affected than LF and HF by performance degradations from metals and liquids	Similar characteristics to UHF but faster read rates. Drawback is microwaves are much more susceptible to performance degradations from metals and liquids
Tag power source	Mainly passive using inductive coupling (near-field)	Mainly passive using inductive coupling (near-field)	Active and passive tags using e-field backscatter in the far field	Active and passive tags using e-field backscatter in the far field
Typical applications	Access control, animal tagging, vehicle immobilisers	Smart cards, access control, payment, ID, item-level tagging, baggage control, biometrics, libraries, laundries, transport, apparel	Pallet and box tagging in supply chain, baggage handling, electronic toll collection	Electronic toll collection, real-time location of goods
Notes	Largest installed base due to mature technology. However, it will be overtaken by higher frequencies	Currently, the most widely available high frequency world-wide due to the adoption of smart cards in transport	Different frequencies and power allocated by different countries: US 4W (EIRP) 915MHz, Europe 0.5W (ERP) 868MHz	5.8 GHz more or less abandoned for RFID
Multiple tag read rate	Slower			Faster
Ability to read near-metal or wet surfaces	Better			Worse
Passive tag size	Larger			Smaller

In a typical system, tags are attached to objects. Each tag has a certain amount of internal memory (EEPROM) in which it stores information

When these tags pass through a field generated by a reader, they transmit this information back to the reader, thereby identifying the object. The antenna uses radio frequency waves

Low-frequency (30-500 kHz) RFID systems have a short transmission range (generally less than 1.8 metres). High-frequency (850-950 MHz and 2.4-2.5 GHz) RFID systems offer a longer transmission range (more than 27 metres). In general, the higher the frequency, the more expensive the system. RFID is sometimes called dedicated short-range communication.

There are two types of RFID tags: read-only tag and read-write tag. In

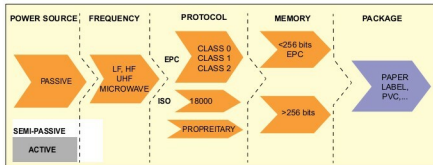


Fig. 2: How passive tags are defined